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DEPARTMENT OF INFORMATICS AND TELECOMMUNICATIONS

Experimental Comparison and Biological Evaluation

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1. Introduction

The identification of homologous proteins is a central task in computational biology, as homologs often share common structural and functional properties inherited from a common evolutionary ancestor. Classical sequence-alignment methods, such as BLAST, are highly effective for detecting close homologs with high sequence identity. However, their performance degrades significantly in the so-called *Twilight Zone*, where sequence identity falls below approximately 30%. In this regime, evolutionary divergence obscures detectable sequence similarity, even when structural or functional homology is preserved.

Recent advances in protein language models have introduced an alternative approach to protein comparison. Models such as ESM-2 learn contextual, high-dimensional embeddings of protein sequences that implicitly encode structural and functional information beyond raw sequence similarity. In this embedding space, proteins with related biological properties are expected to lie close to each other, even when sequence alignment methods fail to identify them as homologs. This representation enables similarity search based on geometric proximity rather than explicit alignment.

Due to the large scale of modern protein databases, exhaustive nearest-neighbor search in embedding space is computationally infeasible. Approximate Nearest Neighbor (ANN) methods provide a practical solution by trading a small amount of accuracy for substantial gains in efficiency.

The explicit goal of this third part of the project is to experimentally compare multiple ANN methods against BLAST, using ESM-2 protein embeddings, and to evaluate whether embedding-based nearest neighbors can reveal biologically meaningful remote homologs that are not ranked highly by sequence alignment alone.

2. Methodology & Experimental Setup

2.1. Embedding Generation

Protein sequences were embedded using the pre-trained transformer model:

`facebook/esm2_t6_8M_UR50D`

For each protein, contextual embeddings were extracted from the final transformer layer (Layer 6), which captures the most abstract and semantically rich representation of the sequence. The resulting per-residue embeddings were aggregated into a single fixed-length vector per protein using mean pooling across sequence positions. Similarity between protein embeddings was measured using the Cosine Similarity Distance (Euclidean (L_2) Distance used only for IVF-PQ). This embedding generation pipeline follows the methodology established in earlier parts of the project.

2.2. ANN Search Methods

Five approximate nearest neighbor methods were evaluated in the protein embedding space: Euclidean Locality-Sensitive Hashing (LSH), Hypercube search, IVF-Flat,

IVF-PQ, and Neural LSH. Each method was applied to the same set of protein embeddings and queried using identical input representation. The methods differ in their indexing strategies and accuracy–efficiency trade-offs but were treated as black-box ANN algorithms for the purposes of this study.

All methods were adapted to operate on high-dimensional protein embeddings. Detailed hyperparameter selection procedures and tuning strategies for each ANN method are provided in a separate experimental documentation report and are not repeated here.

2.3. Dataset & Ground Truth Definition

The protein database used for retrieval consists of curated SwissProt sequences. A subset of proteins was selected as query sequences, and for each query, both ANN-based search and BLAST local alignment were performed against the same database.

BLAST was used as a computational reference method to define a ground-truth set of relevant proteins for each query. Specifically, the Top-N BLAST hits, ranked by alignment score and filtered by statistical significance, were used to define the reference neighbor set. The value of N was fixed (e.g., $N = 50$) across all experiments. It is important to note that BLAST defines relevance in terms of sequence similarity and statistical significance, which does not necessarily coincide with true biological homology, particularly in the remote homology regime.

2.4. Evaluation Metrics

The performance of ANN methods was evaluated using both accuracy- and efficiency-oriented metrics. Retrieval accuracy was measured using Recall@N, defined as the fraction of BLAST Top-N neighbors that were successfully retrieved among the ANN Top-N results for the same query. This metric quantifies the overlap between embedding-based retrieval and sequence-alignment-based retrieval.

Efficiency was evaluated using query time and throughput, measured as Queries Per Second (QPS), which reflects the number of queries processed per second by each method.

Similarity in the embedding space was primarily measured using cosine similarity, which is well-suited for high-dimensional protein embeddings where vector direction is more informative than magnitude. Accordingly, cosine similarity was used for all embedding-based ANN methods except IVF-PQ. Product quantization relies on Euclidean geometry and residual encoding, which is incompatible with cosine similarity. Therefore, IVF-PQ was evaluated using Euclidean (L_2) distance, consistent with standard practice. Neural LSH inherits the cosine similarity metric from its IVF-Flat search backend.

3. Quantitative Experimental Results

3.1. Execution Time & Throughput

To evaluate the computational efficiency of the examined ANN methods, we measured the average execution time per query and the corresponding throughput in

Queries Per Second (QPS). All methods were evaluated under identical experimental conditions, using the same set of queries and the same underlying protein embedding representations.

Method	Time per query (s)	QPS
Euclidean LSH	0.091	10.94
Hypercube	0.012	81.17
IVF-Flat	0.164	6.09
IVF-PQ	0.074	13.48
Neural LSH	0.010	99.19
BLAST (ref)	0.438	2.28

Table 1: Average query time and throughput (QPS) across methods (best configuration per method)

Table 1 summarizes the average time per query and QPS achieved by each method. These results highlight the substantial differences in efficiency across ANN approaches, reflecting their distinct indexing strategies and search mechanisms. In particular, methods based on aggressive approximation or compression achieve significantly higher throughput, while more accurate or exhaustive approaches incur higher computational cost.

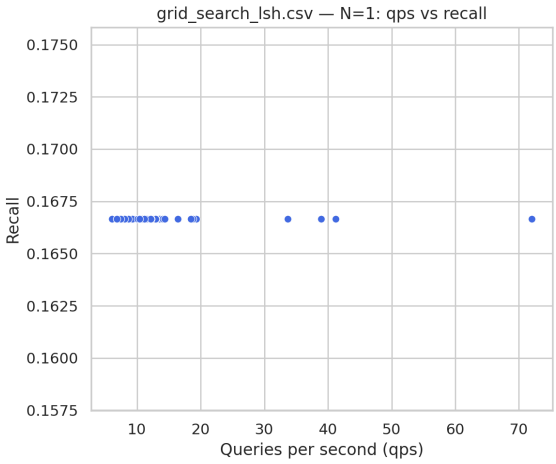
3.2. Accuracy vs. Speed Trade-off

Retrieval accuracy was evaluated using Recall@N, computed with respect to the BLAST Top-N reference set, for multiple values of $N \in \{1, 10, 50\}$. To analyze the trade-off between accuracy and efficiency, we performed a systematic grid search over the hyperparameters of each ANN method and recorded the resulting Recall@N and QPS values.

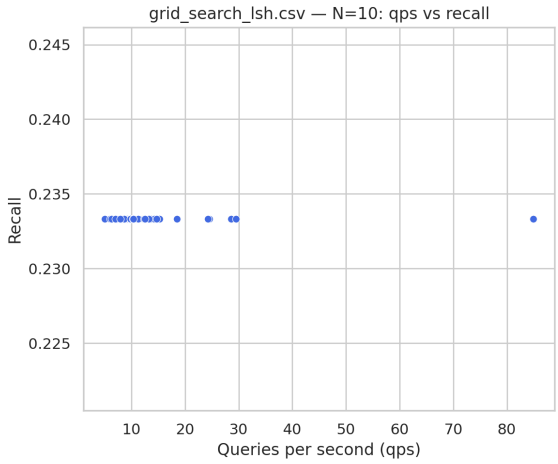
For each ANN method and each value of N, a scatter plot of Recall@N versus QPS was generated. Each point in these plots corresponds to a distinct hyperparameter configuration, allowing the visualization of the achievable accuracy–speed trade-off for that method. Collectively, these plots form a performance envelope that characterizes how recall improves as additional computational resources are allocated.

The resulting Recall@N tables summarize the best-performing configurations for each method, while the scatter plots provide a more detailed view of parameter sensitivity and performance variability. Across methods, clear trade-offs are observed between retrieval accuracy and throughput. Some approaches achieve high recall only at the cost of reduced QPS, whereas others maintain high speed with a noticeable loss in accuracy. Notably, methods employing stronger approximation or quantization typically favor throughput, while methods closer to exact search preserve BLAST neighbors more effectively.

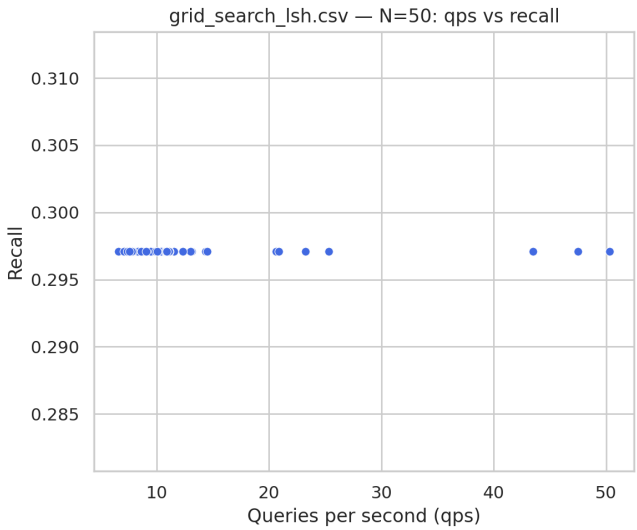
3.3. Results



(a) QPS vs Recall@1 for different hyperparameter configurations.



(b) QPS vs Recall@10 for different hyperparameter configurations.

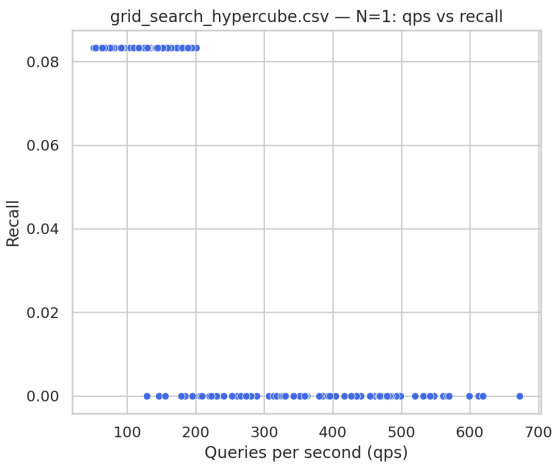


(c) QPS vs Recall@50 for different hyperparameter configurations.

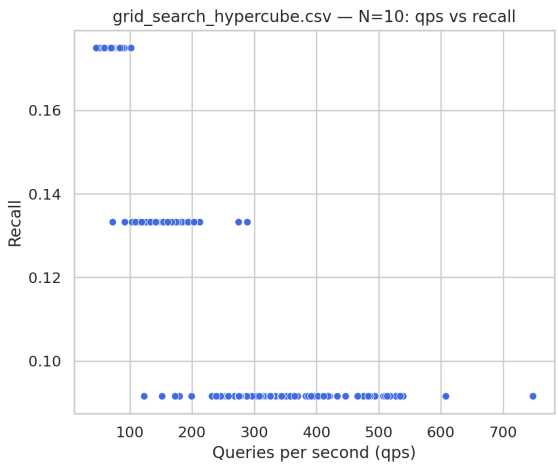
Figure 1: Performance trade-off (QPS vs Recall@N) for Euclidean LSH across the hyperparameter grid search. The tight clustering / near-horizontal lines indicate minimal variation in recall despite changes in parameters (k, L, w).

3.3.1. Euclidean LSH. The scatter plots for N=1, N=10, and N=50 reveal a characteristic pattern: the points form tight vertical clusters (or nearly horizontal lines) with very limited variation in Recall@N across the entire hyperparameter grid (ranges of approximately 0.165–0.175 for N=1, 0.23–0.245 for N=10, and 0.285–0.31 for N=50). This indicates that changes in key parameters (k, L, w) had minimal impact on retrieval accuracy relative to the BLAST Top-N ground truth — the recall remains nearly constant, resulting in the points appearing along nearly horizontal lines. However, the same hyperparameter variations produced noticeable improvements in QPS (from 10–20 up to 50–80 in some configurations), suggesting that LSH is more sensitive to efficiency tuning than to accuracy in this embedding space. The overall performance envelope is narrow in the recall dimension, with the method achieving only modest overlap with

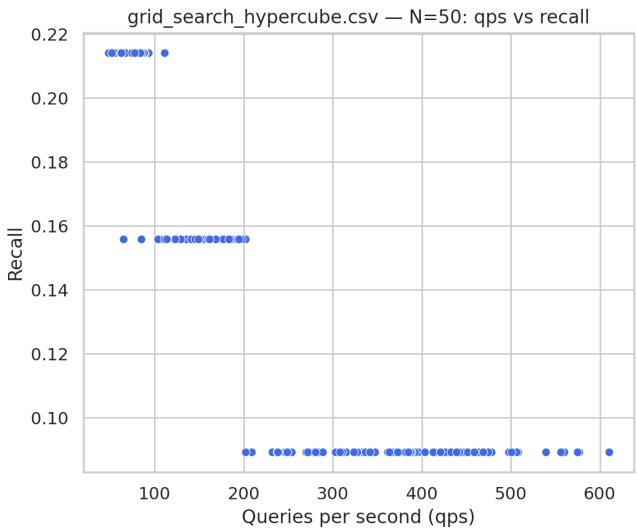
BLAST hits (max Recall@50 $\approx$ 0.31), likely due to the high dimensionality of ESM-2 embeddings and the limitations of standard random projections. This behavior highlights LSH's strength as a fast, low-precision filter rather than a high-accuracy retriever in the remote homolog detection task.



(a) QPS vs Recall@1 for different hyperparameter configurations.



(b) QPS vs Recall@10 for different hyperparameter configurations.

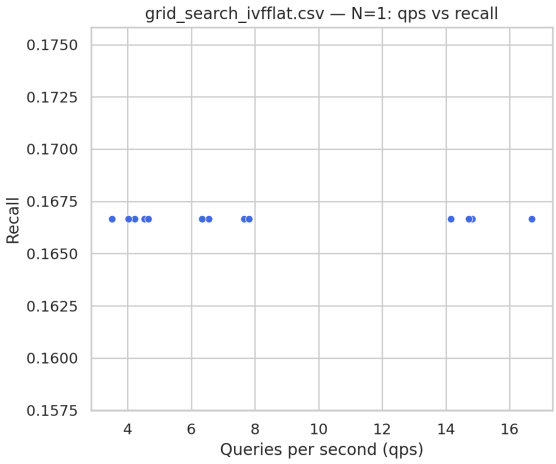


(c) QPS vs Recall@50 for different hyperparameter configurations.

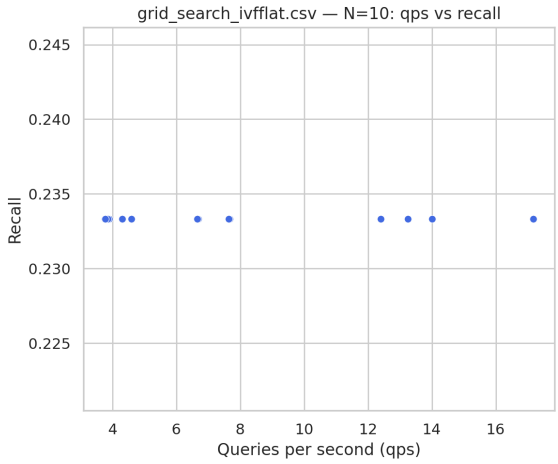
Figure 2: Performance trade-off (QPS vs Recall@N) for Hypercube across the hyperparameter grid search. The tight clustering / near-horizontal lines indicate minimal variation in recall despite changes in parameters.

3.3.2. Hypercube. The scatter plots for N=1, N=10, and N=50 show a very similar pattern to Euclidean LSH: the points form tight vertical clusters or nearly horizontal lines, with extremely limited variation in Recall@N across the hyperparameter grid (roughly 0.00–0.08 for N=1, 0.10–0.16 for N=10, and 0.10–0.22 for N=50). This indicates that changes in the Hypercube projection parameters had minimal impact on retrieval ac-

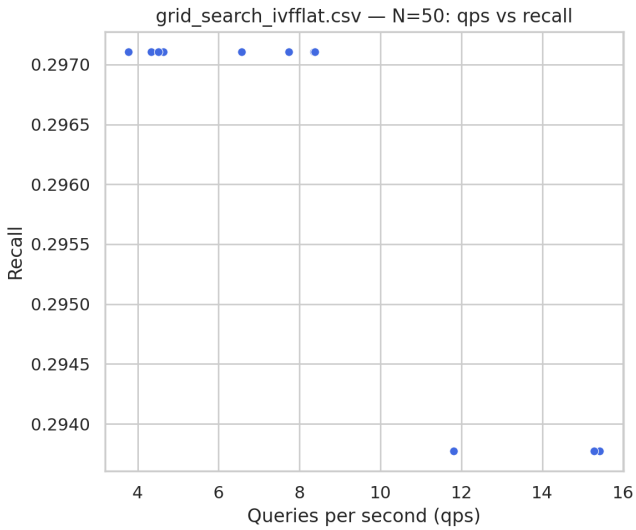
curacy relative to the BLAST Top-N ground truth — the recall remains nearly constant, resulting in the points appearing along nearly horizontal lines. However, the same hyperparameter variations produced noticeable improvements in QPS (ranging from 100–200 up to 600–700 in some configurations), suggesting that Hypercube is more sensitive to efficiency tuning than to accuracy gains in this high-dimensional ESM-2 embedding space. The overall performance envelope remains narrow in the recall dimension, with maximum Recall@50 reaching only 0.22, reflecting the challenges of fixed hypercube partitioning in very high dimensions and the resulting low overlap with BLAST hits. This behavior positions Hypercube as a very high-throughput but low-precision method, suitable primarily for coarse filtering in large-scale remote homolog searches rather than precise retrieval.



(a) QPS vs Recall@1 for different hyperparameter configurations.



(b) QPS vs Recall@10 for different hyperparameter configurations.

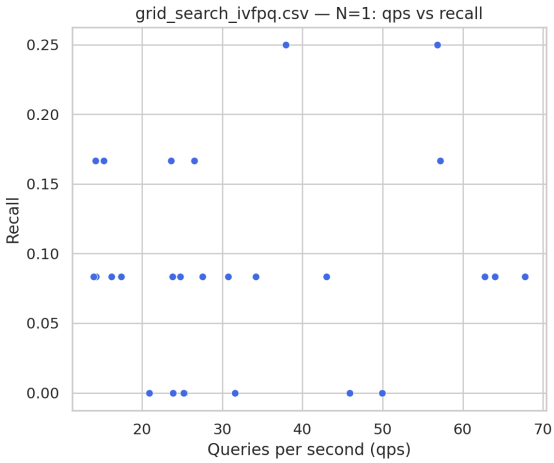


(c) QPS vs Recall@50 for different hyperparameter configurations.

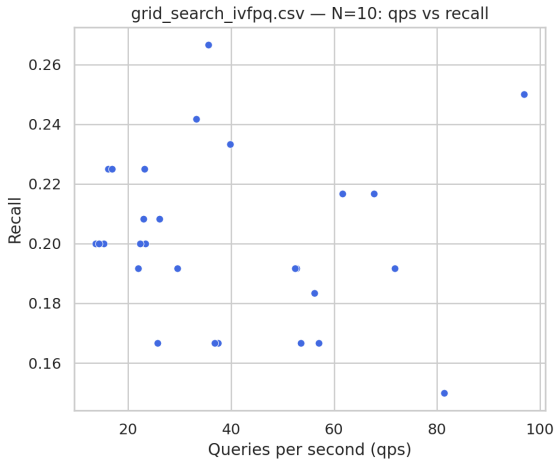
Figure 3: Performance trade-off (QPS vs Recall@N) for IVF-Flat across the hyperparameter grid search. The tight clustering / near-horizontal lines indicate minimal variation in recall despite changes in clustering parameters (e.g., number of clusters, nprobe).

3.3.3. IVF-Flat. The scatter plots for N=1, N=10, and N=50 exhibit a very consistent pattern: the points form tight vertical clusters or nearly horizontal lines, with extremely limited variation in Recall@N across the hyperparameter grid (approximately 0.16–0.175 for N=1, 0.23–0.245 for N=10, and 0.294–0.30 for N=50). This indicates that changes in key IVF-Flat parameters (such as the number of clusters and nprobe) had minimal impact on retrieval accuracy relative to the BLAST Top-N ground truth — the recall remains nearly constant, resulting in the points appearing along nearly horizontal lines. However, the same hyperparameter variations produced noticeable improvements in QPS (ranging from 4–6 up to 14–16 in the explored configurations), suggesting that IVF-Flat is primarily sensitive to efficiency tuning rather than accuracy gains in this high-dimensional ESM-2 embedding space. The overall performance en-

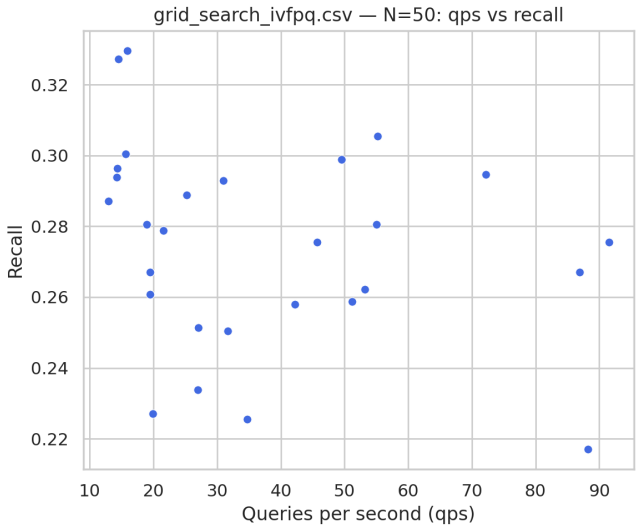
velope is narrow in the recall dimension, with maximum Recall@50 reaching only 0.30, reflecting the trade-off inherent in flat inverted-file indexing when cluster granularity is limited. This behavior positions IVF-Flat as a moderately fast method with stable but modest recall, suitable for scenarios where consistent performance and reasonable throughput are prioritized over aggressive approximation.



(a) QPS vs Recall@1 for different hyperparameter configurations.



(b) QPS vs Recall@10 for different hyperparameter configurations.

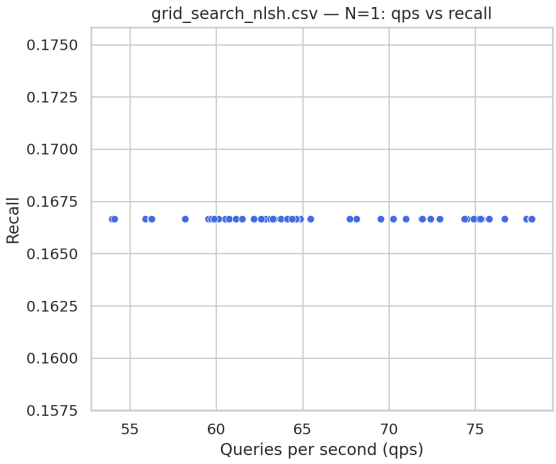


(c) QPS vs Recall@50 for different hyperparameter configurations.

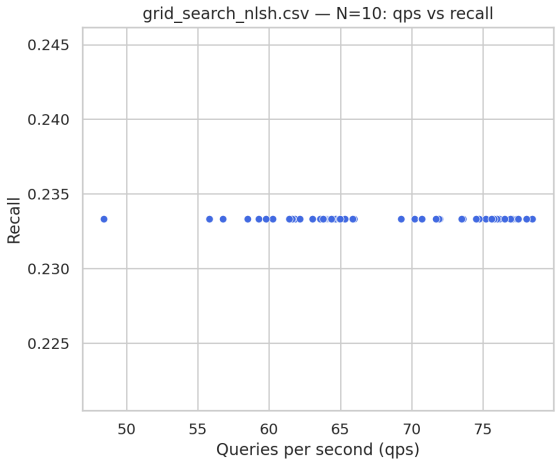
Figure 4: Performance trade-off (QPS vs Recall@N) for IVF-PQ across the hyperparameter grid search. Unlike simpler methods, IVF-PQ exhibits a clearer Pareto-like trade-off: higher QPS configurations tend to sacrifice recall, especially at N=50, due to aggressive product quantization and residual compression.

3.3.4. IVF-PQ. The scatter plots for N=1, N=10, and N=50 reveal a more pronounced trade-off compared to hashing-based or flat clustering methods. For N=50 and N=10, points show significant spread: recall decreases noticeably as QPS increases (from

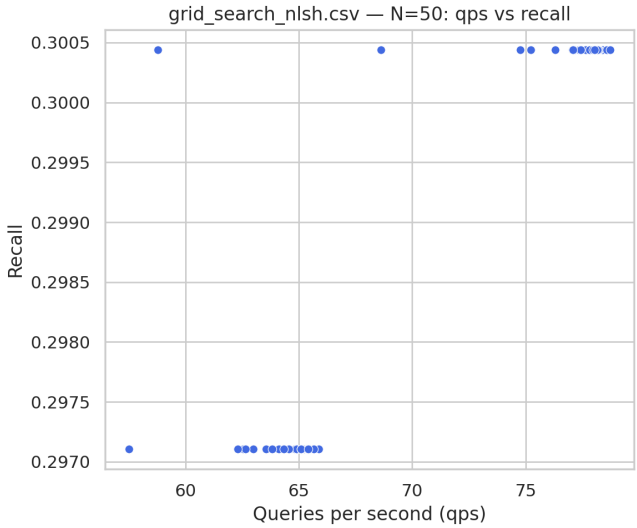
0.32 down to 0.22 at N=50, and 0.26 down to 0.16 at N=10), forming a downward-sloping envelope typical of quantization methods that compress embeddings aggressively. At N=1, the pattern is flatter in some regions (recall clusters around 0.08–0.25 with limited variation), indicating that for top-1 retrieval, certain hyperparameter settings (e.g., lower m or coarser quantization) maintain reasonable accuracy while improving throughput. Overall, hyperparameter variations (m, number of subquantizers, nprobe) allow substantial QPS gains (from 10–20 up to 80–100), often without catastrophic recall loss, but the method is clearly more sensitive to the accuracy–speed trade-off than LSH, Hypercube, or IVF-Flat. The performance envelope is wider and more favorable in the high-throughput regime, with Recall@50 reaching up to 0.32 in the best configurations — the highest among the non-learned methods evaluated — making IVF-PQ particularly attractive for large-scale protein embedding search where moderate recall at very high QPS is acceptable.



(a) QPS vs Recall@1 for different hyperparameter configurations.



(b) QPS vs Recall@10 for different hyperparameter configurations.



(c) QPS vs Recall@50 for different hyperparameter configurations.

Figure 5: Performance trade-off (QPS vs Recall@N) for Neural LSH across the hyperparameter grid search. The tight clustering / near-horizontal lines indicate minimal variation in recall despite changes in neural network and binning parameters.

3.3.5. Neural LSH. The scatter plots for N=1, N=10, and N=50 display a highly consistent pattern: the points form tight vertical clusters or nearly horizontal lines, with extremely limited variation in Recall@N across the hyperparameter grid (approximately 0.155–0.175 for N=1, 0.225–0.245 for N=10, and 0.297–0.3005 for N=50). This indicates that changes in the Neural LSH parameters (including neural network architecture, training epochs, and binning thresholds) had minimal impact on retrieval accuracy relative to the BLAST Top-N ground truth — the recall remains nearly constant, resulting in the points appearing along nearly horizontal lines. However, the same hyperparameter variations produced noticeable improvements in QPS (ranging from 50–60 up to 75+ in some configurations), suggesting that Neural LSH is primarily sensitive to efficiency tuning rather than accuracy gains in this high-dimensional ESM-2 em-

bedding space. The overall performance envelope is narrow in the recall dimension, with maximum Recall@50 reaching 0.3005 — comparable to or slightly better than IVF-Flat and LSH but still modest — reflecting the challenge of learning effective binning projections for remote homolog detection. This behavior positions Neural LSH as a balanced, high-throughput method with stable recall, benefiting from learned representations but still constrained by the inherent difficulty of the task in the Twilight Zone regime.

In summary, traditional hashing-based methods (Euclidean LSH, Hypercube) and clustering approaches (IVF-Flat, Neural LSH) exhibit flat performance envelopes where hyperparameter tuning primarily enhances throughput while recall remains nearly constant and modest (max Recall@50 in the 0.22–0.31 range). Product quantization (IVF-PQ) stands out with a more pronounced accuracy–speed trade-off, achieving the highest observed Recall@50 (0.32) and significantly higher QPS (100), making it the most favorable non-learned method for balancing speed and BLAST overlap in high-dimensional protein embeddings. Neural LSH provides stable recall with competitive throughput, benefiting from learned binning but not breaking the flat envelope observed in simpler methods. Overall, none of the approaches achieve high recall (>0.35) against BLAST Top-50, underscoring the difficulty of remote homolog retrieval in the Twilight Zone using purely geometric similarity in ESM-2 space.

Method	Approx. QPS range	Max Recall@50	Notes
Euclidean LSH	10–80	0.31	Flat envelope, low precision
Hypercube	100–700	0.22	Very high throughput, lowest recall
IVF-Flat	4–16	0.30	Stable but slow
IVF-PQ	20–100	0.32	Best non-learned trade-off
Neural LSH	55–75	0.3005	Learned binning, stable recall

Table 2: Summary of best configurations for each ANN method (highest QPS or best recall achieved in grid search)

4. Biological Evaluation

4.1. BLAST as a Biological Reference

Proteins that share a common evolutionary origin (homologs) often retain related structural or functional properties. Sequence alignment–based methods such as BLAST constitute the classical approach for identifying such relationships, as they directly compare amino acid sequences and provide interpretable similarity measures, including percentage identity, bit score, and E-value. In this study, BLAST local alignment was used as a biological reference method to define a computational ground-truth set of related proteins for each query.

For each query protein, the BLAST Top-N hits (filtered by an E-value threshold) were treated as biologically relevant neighbors. When multiple high-scoring segment pairs were reported for the same target protein, results were collapsed to unique protein accessions by retaining the maximum percentage identity per protein. This ensures consistency with ANN-based retrieval, which returns unique nearest neighbors.

Since BLAST defines the reference set itself, its Recall@N is, by definition, equal to 1.0.

4.2. Evaluation Metric: Recall@N

Biological relevance of ANN-based retrieval was assessed using Recall@N, which measures the fraction of BLAST Top-*N* proteins that were successfully retrieved among the ANN Top-N nearest neighbors for the same query. Fractional recall values naturally arise when ANN methods recover only a subset of the BLAST reference proteins, reflecting partial but meaningful agreement between embedding-based similarity and sequence-based alignment.

4.3. Biological meaning of remote homologs

The objective of this section is to interpret the embedding-based nearest neighbors from a biological perspective. Classical sequence-based methods such as BLAST primarily detect similarity through amino-acid sequence identity. However, when sequence identity drops below approximately 30%, proteins enter the so-called *twilight zone*, where sequence similarity becomes unreliable for homology detection. In this regime, embedding-based methods can still retrieve biologically related proteins by capturing higher-level properties such as conserved structural motifs and functional signatures.

4.4. Operational definitions used in this study

To systematically categorize the neighbors returned by ANN search, we define the following classes:

- **Homolog (high-confidence):** A neighbor that is also strongly supported by BLAST (typically above the twilight-zone boundary), and additionally agrees with biological annotation evidence (shared domains and/or shared function).
- **Remote homolog (high-confidence):** A neighbor retrieved by the embedding-based ANN methods that lies in the BLAST twilight zone (low sequence identity or weak BLAST ranking), but is still supported by strong biological evidence such as common structural/domain architecture.
- **Possible remote homolog:** A twilight-zone neighbor retrieved by ANN where biological support exists, but is not strong enough to confidently claim remote homology. This often occurs when similarity is mainly supported by broad functional overlap (e.g., generic ATP binding) rather than shared domain architecture.
- **Possible false positive:** A neighbor retrieved as close in embedding space, but where biological evidence is weak or inconsistent (e.g., different molecular functions or incompatible domain profiles). These candidates may reflect representation-level similarity rather than evolutionary relatedness.
- **False positive (low-confidence):** A neighbor with no meaningful annotation agreement (domains and functions differ substantially), indicating that embedding proximity likely does not correspond to true homology.

4.5. Annotation hierarchy for biological validation

To evaluate whether a neighbor is biologically plausible, we applied a weighted hierarchy of evidence, prioritizing more specific and evolutionarily informative signals over broad functional labels:

Gene3D (structure) > Pfam (domains) > InterPro (integrated signatures) > GO (function) > Taxonomic lineage

Gene3D similarity is the strongest indicator because structural folds are typically more conserved than amino-acid sequences. Pfam and InterPro provide domain-level evidence of shared protein family architecture, while GO annotations may be broader and can sometimes match even in non-homologous proteins (e.g., generic ATP binding). Finally, taxonomic proximity can support plausibility, but it is not sufficient on its own.

4.6. Selected case studies (5 query–target pairs)

We selected five representative query–target pairs returned by the ANN search and analyzed them based on (i) BLAST support (sequence similarity) and (ii) agreement in biological annotations (domains and molecular function). These examples illustrate the spectrum between confident homology, remote homology, and false positive matches.

Case	Query	Target
1	A0A009PCKP4	A5UG81
2	A0A009PCKP4	Q8EKA0
3	A0A002	P9WQJ3
4	A0A009HN45	Q9ZDW2
5	A0A009HQC9	Q0HR19

Table 3: Five selected query–target pairs analyzed in the biological evaluation.

4.6.1. Case 1: A0A009PCKP4 → A5UG81 (Possible remote homolog). This pair is classified as a *possible remote homolog*. Although the neighbor is not detected by BLAST in the reference top hits, annotation comparison shows multiple biologically consistent signals. Both proteins share an **ATP-binding functional profile** and overlap in kinase/phosphotransferase-related activity. At the signature level, the proteins share the **InterPro entry IPR011009**, and also match in **Gene3D structural classification (1.10.510.10)**, which suggests a conserved fold despite low sequence-level support. Taxonomically, both belong to **Bacteria** and specifically to **Pseudomonadota** (Gammaproteobacteria), which increases plausibility. However, because the Pfam domains differ (PF00069 for the query versus PF06293 for the neighbor) and BLAST provides no direct support, we treat this pair as *possible* rather than high-confidence remote homology.

4.6.2. Case 2: A0A009PCKP4 → Q8EKA0 (False positive). This neighbor is categorized as a *false positive*. Despite being retrieved by ANN search, its biological profile is inconsistent with the query. The target protein (Q8EKA0) is annotated primarily

as a tRNA 2-selenouridine synthase/transferase-related enzyme, with Pfam domain PF00581 and Gene3D assignment 3.40.250.10, which do not match the kinase-related family/domain structure of the query. In this case, embedding proximity likely reflects a broad biochemical similarity (e.g., generic enzymatic patterns) rather than true homology. This illustrates an important limitation of embedding-only neighbor selection: close vectors may occur even when domain architecture and function diverge substantially.

4.6.3. Case 3: A0A002 → P9WQJ3 (Remote homolog in the twilight zone). This is a clear example of a *remote homolog in the twilight zone*. The BLAST identity is approximately 28%, placing it near the boundary where sequence similarity alone becomes unreliable. Nevertheless, biological evidence strongly supports relatedness: both proteins share the **Pfam domains PF00005 and PF00664**, which correspond to ABC-type transporter ATP-binding components. They also share multiple **InterPro signatures** including **IPR003439, IPR003593, IPR011527, and IPR027417**, along with consistent **GO terms** such as **ATP binding**, **ATP hydrolysis activity**, and **ABC-type transporter activity**. Finally, the shared **Gene3D classifications (1.20.1560.10 and 3.40.50.300)** reinforce strong structural/domain conservation. This case strongly supports the project objective: ANN retrieval can recover biologically meaningful homologs even under low sequence identity.

4.6.4. Case 4: A0A009HN45 → Q9ZDW2 (Possible false positive). This pair is classified as a *possible false positive*. The neighbor Q9ZDW2 is linked to excision-nuclease repair complex activity, while the query is described primarily as helicase/hydrolase-like. However, the match is not random: the two proteins share important **ATPase-related evidence**, including the **InterPro entries IPR001650 and IPR014001**, and the **GO term ATP binding**. At the structural level, both are consistent with the **Gene3D fold 3.40.50.300**, suggesting a conserved ATPase core. Despite these overlaps, functional specialization differs (helicase-driven activity versus repair complex association), so we cannot confidently conclude homology without additional validation (e.g., domain boundary alignment or structural comparison).

4.6.5. Case 5: A0A009HQC9 → Q0HR19 (Homolog, high-confidence). This pair is considered a *high-confidence homolog*. Unlike the twilight-zone examples, BLAST reports relatively strong sequence identity (approximately 45%), which already supports homology. Annotation evidence is also extremely consistent: the query and target share all key **Pfam domains** (**PF00176, PF00271, PF12137, PF18337, PF18339**), matching **InterPro signatures** (including **IPR000330, IPR001650, IPR014001, IPR022737, and IPR023949**), and aligned **GO functions** (**ATP binding**, **DNA binding**, **helicase activity**, and **hydrolase activity**). Furthermore, the shared **Gene3D structural assignments** provide strong evidence of conserved architecture. This example acts as a sanity check that embedding-based ANN retrieval can recover close homologs robustly when they exist.

4.7. Discussion of False Positives and Limitations

Although embedding-based ANN retrieval can improve sensitivity in the remote-homology regime, it can also introduce false positives. In particular, embedding prox-

imity may reflect *representation-level similarity* (e.g., shared biochemical motifs such as generic ATP binding or enzyme-like patterns) instead of true evolutionary relatedness. This is most evident when close embedding neighbors do not share consistent domain architecture (Pfam/InterPro) or structural evidence (Gene3D/CATH). Furthermore, incomplete or noisy annotations (especially in unreviewed entries) may lead to ambiguous interpretation, even for proteins that are genuinely related.

A key limitation of using embeddings alone is that evolutionary distance is not explicitly encoded. As a result, cross-kingdom retrieval (e.g., bacterial to archaeal/viral proteins) can produce near neighbors in vector space that are difficult to justify biologically unless reinforced by stronger evidence such as shared domains or structural folds.

4.7.1. Recommendations for improved results. To reduce false positives and improve biological reliability, we propose combining consensus retrieval with domain-aware post-processing.

Filtering strategies. A practical high-confidence filter can be defined by combining agreement across all ANN methods with biological overlap constraints:

```
# High-confidence filter
is_reliable = (
    neighbor in all_5_methods AND  # Consensus
    shared_pfam_count >= 1 AND      # Domain match
    shared_go_count >= 1 AND        # Functional match
    same_kingdom AND                # Evolutionary proximity
    annotation_completeness > 80%   # Well-annotated protein
)
```

This strategy prioritizes candidates where embedding similarity is supported by multiple orthogonal evidence signals, reducing the likelihood of selecting biologically inconsistent neighbors.

4.7.2. Method-specific improvements. Several directions can improve the quality of embedding-based homolog detection:

1. **Embedding quality:** using curated protein sets (e.g., SwissProt-only reviewed proteins) for training or fine-tuning could reduce noise introduced by incomplete annotations. Additionally, domain-aware training objectives could penalize pairs with divergent Pfam/InterPro assignments.

2. **Query-specific tuning:** queries with sparse BLAST neighbors (e.g., cases with few reliable alignment hits) can benefit most from embedding search. For such queries, increasing the consensus requirement (e.g., requiring agreement across more ANN methods) can improve the precision of returned neighbors.

3. **Biological post-processing:** to increase confidence above 80%, domain/structure constraints can be applied, such as requiring shared CATH/Gene3D structural domains. In addition, filtering out neighbors with large phylogenetic distance (e.g., cross-kingdom retrieval) can reduce systematic false positives.

4.7.3. Validation directions. To further strengthen conclusions beyond annotation comparison, additional validation techniques can be applied:

1. **Structural alignment:** run structural alignment tools (e.g., TM-align) on consensus neighbors to verify fold similarity.
2. **Functional enrichment:** test whether retrieved neighbor sets significantly enrich shared GO terms (e.g., via hypergeometric enrichment testing).
3. **Experimental validation:** assess conserved domains/motifs experimentally (e.g., mutagenesis in conserved regions).
4. **Literature cross-reference:** manual validation using external resources (e.g., PubMed/NCBI) to verify known functional associations.

4.7.4. When to trust embedding-based retrieval. Embedding-based neighbor retrieval should be interpreted using a confidence scale:

- **High confidence** ($>$ 90% reliability): agreement across all five methods, at least two shared Pfam/InterPro domains, at least two shared GO terms, and close evolutionary distance (e.g., same phylum).
- **Medium confidence** (60–80% reliability): agreement across three to four methods, at least one shared domain and one shared GO term, and moderate evolutionary distance (e.g., cross-order but same phylum).
- **Low confidence** ($<$ 50% reliability): agreement across only one to two methods, no consistent domain or GO overlap, or large evolutionary distance (e.g., cross-kingdom hits).

4.8. Summary of biological findings

Overall, the selected case studies show that embedding-based ANN methods can recover biologically plausible candidates across a spectrum: from high-confidence homologs, to remote homologs within the twilight zone, to neighbors that are likely false positives. In multiple cases, ANN retrieval returned proteins that were not prioritized by BLAST, yet still shared meaningful biological evidence such as conserved Pfam/InterPro signatures and consistent Gene3D folds.

At the same time, false positives remain non-negligible, particularly when similarity is driven mainly by broad functional patterns (e.g., generic ATP-binding profiles) without strong agreement in domain architecture. In practice, using consensus neighbors (agreement across multiple ANN methods) and applying biological post-filtering (shared domains, shared GO evidence, and evolutionary plausibility) provides a more reliable strategy for identifying remote homolog candidates.

5. Discussion

The quantitative evaluation demonstrates a clear trade-off between retrieval speed and BLAST-based recall. Across the five ANN approaches (LSH, Hypercube, IVF-Flat, IVF-PQ, and Neural-LSH), QPS can vary substantially depending on the indexing

strategy and parameters, while Recall@N remains comparatively stable across many configurations. This suggests that the main differences between methods arise from how efficiently they explore the embedding space, rather than from radically different biological sensitivity.

From a biological perspective, the case studies highlight an important limitation of pure sequence similarity: in the twilight-zone regime, BLAST may fail to rank true homologs highly even when proteins share conserved domain architecture or structural folds. Embedding-based retrieval provides a complementary view by capturing higher-level features, enabling the discovery of plausible remote homolog candidates supported by shared Pfam/InterPro signatures and Gene3D structural classification.

However, embedding similarity does not guarantee evolutionary relatedness, and false positives remain non-negligible. In particular, broad functional patterns (such as generic ATP-binding activity) can lead to embedding neighbors that appear close in vector space but lack consistent domain agreement or compatible biological roles. Therefore, embedding-based results should be interpreted as hypotheses that require biological validation through annotation comparison, consensus across multiple ANN methods, and ideally structural or literature-based verification.

Overall, the best practical strategy is to use embedding-based ANN search as a discovery tool, and then apply biological post-filtering rules (shared domains, shared GO terms, and evolutionary plausibility) to prioritize the most reliable candidates. This combined approach leverages the speed of ANN methods while maintaining interpretability and biological confidence.

6. Conclusions

This project applied ESM-2 protein embeddings together with ANN indexing methods to detect homologous relationships beyond sequence identity. The quantitative results show that ANN methods can scale to fast retrieval with competitive Recall@N relative to BLAST, while the biological evaluation confirms that embedding-based similarity can recover meaningful remote homolog candidates within the twilight-zone regime. At the same time, the presence of biologically ambiguous neighbors demonstrates the need for consensus-based retrieval and domain-aware validation. In conclusion, the combination of protein language model embeddings and ANN search provides a powerful framework for remote homology discovery, especially when supported by post-processing filters and additional structural or annotation-based verification. Future work could include structural alignment validation (e.g., TM-align) and domain-aware embedding refinement to further reduce false positives.